

William Bumgardner

Curriculum Producer — AI Curriculum, Content Systems, and Production Tooling

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SUMMARY

I build and ship curriculum at volume, and I build the tooling that keeps the output consistent. Five complete technical courses authored end to end, each with day-by-day modules, learner guides, assessments, and instructor materials. Alongside them I build the AI infrastructure the production runs on: agentic skills that execute documented workflows the same way every time, automated integrity checks on the content library, and pipelines that turn one finished unit into multiple learner-facing formats. My instinct is to notice the workflow I have repeated three times and turn it into a system, then leave it documented enough that someone else can run it.

SELECTED BUILDS

Agentic skills that codify repeatable production work

Authored a suite of documented AI skills that trigger on intent and execute a defined workflow end to end. Each encodes a named protocol, a preflight check, and a done-means checklist rather than relying on memory. The research-compile skill is the clearest example: I noticed that a large share of my source files were empty stubs, and that compiling them silently produced authoritative-sounding content synthesized from model knowledge rather than the actual source. I extracted that failure into an anti-fabrication gate that classifies every source before processing, halts on thin material instead of guessing, and requires explicit provenance labeling when compiling from general knowledge. The failure mode can no longer happen quietly.

Standards Crosswalk Auditor — live at willbumgardner.com/curriculum-portfolio

Python and JavaScript application that parses day-by-day content against a parsed standards catalog and reports genuine coverage gaps while filtering scope mismatches that would otherwise register as false positives. Lesson content is the single source of truth and regeneration runs through a script, so reports cannot drift from source material. It found that a course I had been reporting at 77 percent coverage was actually at 54 percent. I re-verified the finding by hand, then rebuilt all 17 units against it.

Library health as automation, not discipline

Wired automated integrity checks into the content library that flag broken cross-references, orphaned assets, and modules missing required sections, running on save rather than as a periodic cleanup. Maintain a master index plus per-domain indices so any asset is findable directly, and a registry that prevents the same source from being processed twice.

Single-source content pipeline

Built a pipeline that converts one finished unit into a study guide, flashcards, a self-check quiz, a mind map, and an audio overview without re-authoring the source. Teacher-only material including answer keys and rubrics is excluded by default, and a dry-run step prints and confirms the exact source list before anything generates, so the review gate lives inside the tool.

Scheduled automation in daily use

Standing scheduled jobs that run without supervision: weekly intake routing, a read-only system health audit against a baseline allowlist, and a citation verifier that checks references against CrossRef to catch fabricated or garbled sources. Each is documented and hand-off ready.

EXPERIENCE

Curriculum Developer and Computer Science Instructor

Dayton High School, Lyon County School District, NV | 2019 to Present

- Authored five complete technical courses from nothing: AP Computer Science Principles, Advanced Computer Science I and II, Digital Game Development, and Web Design and Development. Each ships with day-by-day modules, learner guides, formative and summative assessments, and instructor facilitation materials.
- Aligned all content to three external standards frameworks (AP Computer Science Principles, CSTA, Nevada CTE) and documented coverage at the individual lesson level.
- Built reusable unit templates and a documented authoring workflow so each new build starts from a validated structure and quality stays consistent across five distinct subjects.
- Migrated an entire course pathway off a legacy commercial platform onto open tooling, rewriting every activity and assessment.
- Designed and facilitated district-wide professional development for adult learners in AI literacy, data literacy, and educational technology.

- Build the AI tooling the work runs on, using agentic coding tools daily against a high accuracy bar.

Academic Intervention Instructor

Dayton High School, Lyon County School District, NV | 2014 to 2019

EDUCATION

PhD, Computer Science

National University | expected 2028

EdS, Curriculum and Instruction, Computer Science emphasis

University of Florida | 2025

CREDENTIALS

- AP Computer Science Principles Exam Reader, College Board, 2020 to present. Annual calibration and scoring against a standardized national rubric.
- Secondary instructor representative, Nevada Digital Game Development Standards Committee, 2019 to 2020.
- Nevada 7-12 Computer Science Teaching Credential.

TOOLS

Claude and Claude Code, agentic AI workflows, Claude Skills authoring, prompt design · Python · JavaScript · HTML/CSS · Git and GitHub Pages · Markdown-based content systems, structured indices, and linked knowledge bases · shell automation and scheduled jobs